

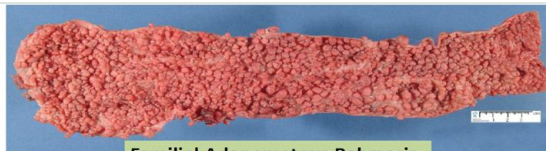
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What gene is mutated in FAP, and what is its normal function?
2. How does the clinical presentation of right-sided vs left-sided colorectal carcinoma differ?
3. What cell of origin gives rise to GIST, and where in the GIT is it most common?

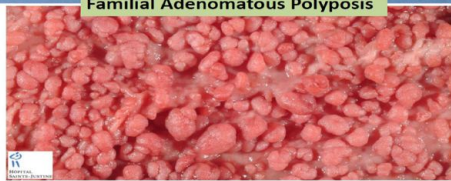
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Familial adenomatous polyposis (FAP)

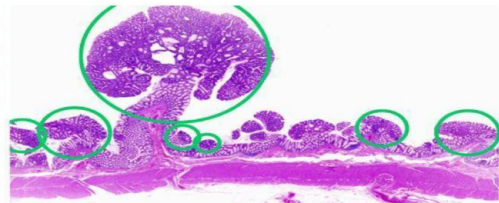
What	Autosomal dominant disorder from mutation of the APC gene (chromosome 5, long arm).
Mechanism	APC is a tumor suppressor gene ; its product regulates cell proliferation via the WNT signaling pathway.
Marked by	Numerous colorectal adenomas appearing by the teenage years ; ≥100 polyps needed for a diagnosis of classic FAP (as many as several thousand may be present).
Malignant potential	Colorectal adenocarcinoma develops in 100% of untreated FAP , often before age 30.
Treatment	Prophylactic colectomy in the late 2nd decade is standard therapy for APC-mutation carriers.
Still at risk of	Extracolonic neoplasia at other sites, even after colectomy.



Familial Adenomatous Polyposis



FAP: colon studded with innumerable adenomatous polyps



Familial Adenomatous Polyposis
Morphologically indistinguishable from sporadic adenomas.

FAP histology: morphologically indistinguishable from sporadic adenomas

2 Intestinal malignancy — where it arises

Site	Frequency
Large intestine (LI)	Commonest GIT malignancy site (adenocarcinoma)
Small intestine (SI)	Very uncommon compared with LI
Exception: peri-ampullary duodenum	Adenomatous lesions around the ampulla of Vater are well recognized

SI and LI adenomas/adenocarcinomas have **similar microscopic features**.

3 Colorectal carcinoma (CRC)

Histologic type	Vast majority are adenocarcinoma.
Epidemiology	Incidence peaks at 60–70 years ; males affected slightly more than females.
Precursor	Most CRC arise in adenomatous polyps.

Dietary risk factors

Factor	Proposed mechanism
Low fiber / high refined carbohydrate	Decreased stool bulk and altered gut microbiota → increased toxic oxidative bacterial by-products
High fat intake	Increases hepatic cholesterol and bile-acid synthesis, which bacteria convert to carcinogens
Vitamin A, C, E deficiency	Loss of antioxidant effect

Aspirin/NSAIDs are protective — likely via **COX-2 inhibition** (COX-2 makes prostaglandin E2, which promotes epithelial proliferation).

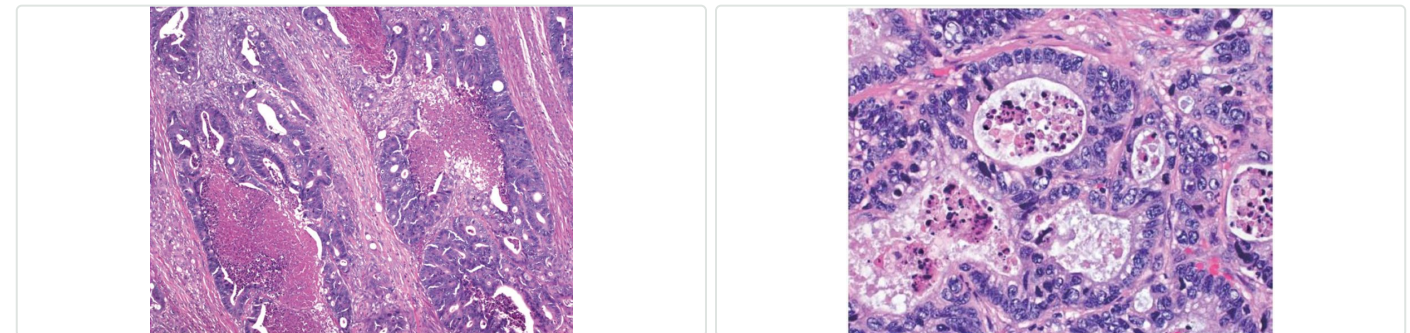
Pathology and clinical features

Adenocarcinoma histology: malignant cells in a **glandular pattern**, with marked **necrosis** and infiltration of the muscle layer.

Location	Presentation
Right colon	Polypoid, fungating mass · presents as insidious anaemia
Left colon	Circumferential, infiltrating “ napkin-ring ” constriction · bleeding PR, change in bowel habit, obstruction
Rectum	Ulcerated, raised plaques · fresh bleeding PR, tenesmus, obstruction



Clinical presentation by tumor site: right colon, left colon, rectum



Adenocarcinoma: glandular pattern with dirty necrosis, invading the muscle wall

Adenocarcinoma, high power: malignant glands with necrotic luminal debris

Spread of CRC

Route	Detail
Direct	Through the intestinal wall
Lymphatic	Into regional lymph nodes
Blood	Liver (via portal vein) · lungs, bone, brain (via systemic veins)
Transcoelomic	Across the peritoneal cavity

4 Neuroendocrine tumor (NET) — illness script

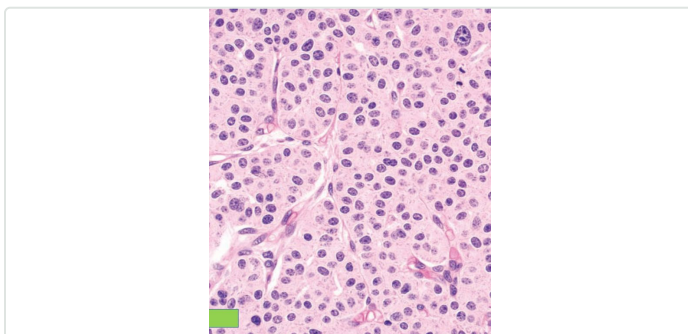
WHO/WHAT	Arises from the diffuse endocrine system (neural crest derivatives). Most common site: midgut-derived epithelium — ileum and appendix. Historically called “carcinoid”; “ neuroendocrine cell tumor ” is now preferred.
Behavior	All have malignant potential , but many are indolent. Ileal NET = low-grade, may metastasize to nodes/liver. Appendiceal NET tend to behave benignly .
Local effects	Mass effect → intestinal obstruction, intussusception .
Systemic effects	Foregut NET (incl. pancreatic) → gastrin, insulin. Midgut NET → serotonin (5-HT) , inactivated by the liver to 5-HIAA (raised urinary 5-HIAA aids diagnosis).
Carcinoid syndrome	Flushing, diarrhea, right-sided endocardial/valvular thickening (tricuspid and pulmonary stenosis); occurs when liver metastases release 5-HT directly into the systemic circulation, bypassing hepatic inactivation.

Pathology of NET

Gross	Intramural/submucosal polypoid masses; yellow-tan , very firm.
Grading	Graded 1–3 by differentiation (atypia) and mitotic activity.
Microscopic (well-differentiated)	Islands, trabeculae, strands, glands or sheets of uniform cells with scant pink granular cytoplasm and a round-to-oval stippled nucleus .



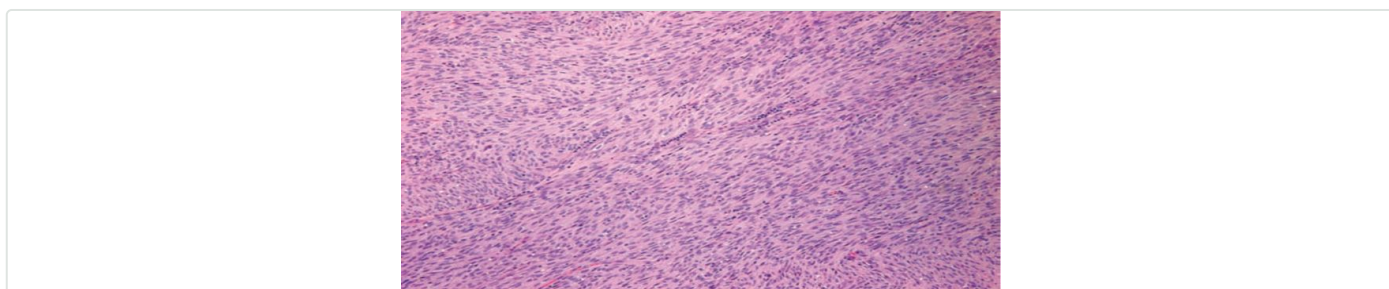
NET: yellow-tan, firm, submucosal polypoid nodules



NET histology: uniform cells with stippled nuclei in nests

5 Gastrointestinal stromal tumor (GIST)

What	The most common mesenchymal tumor of the GIT ; believed to arise from the interstitial cells of Cajal (the GIT's “pacemaker cells” that coordinate peristalsis).
Where	Any part of the GIT; most common in the stomach (60%) , then small bowel (30%) .
Presentation	Nonspecific: pain, GI bleeding, mass effect, obstruction — depends on location and size.
Behavior/risk	Predicted by size, anatomic location, and mitotic rate → classified very low / low / intermediate / high risk. Higher risk = more recurrence/metastasis .
Pathology	Findings
Gross	Usually a well-circumscribed mass; sometimes polypoidal
Microscopic	70% spindle cell · ~20% epithelioid · ~10% mixed spindle + epithelioid



GIST: spindle cell morphology (predominant pattern)

P Must-remember pearls

- **FAP = APC gene mutation (chr 5, tumor suppressor, WNT pathway);** ≥ 100 polyps for diagnosis; **100% progress to CRC** if untreated $\rightarrow$ prophylactic colectomy.
- **Right colon CRC $\rightarrow$ anaemia (polypoid mass); left colon CRC $\rightarrow$ napkin-ring obstruction; rectal CRC $\rightarrow$ bleeding + tenesmus.**
- CRC spreads directly, lymphatically, hematogenously (**liver via portal vein** first), and transcoelomically. Aspirin/NSAIDs protect via **COX-2 inhibition**.
- **NET = neural crest derived, midgut/ileum & appendix most common.** Ileal NET behaves low-grade malignant; appendiceal NET is usually benign. **Carcinoid syndrome needs liver mets** to bypass hepatic 5-HT inactivation.
- **GIST arises from interstitial cells of Cajal**, most often in the **stomach**; 70% are spindle cell; behavior predicted by size, location, mitotic rate.

Q Test yourself

1. A teenager is found to have over 100 colorectal adenomatous polyps on colonoscopy. Genetic testing would most likely reveal a mutation in:

- A. APC gene
- B. KRAS gene
- C. TP53 gene
- D. MLH1 gene
- E. BRCA1 gene

2. A 65-year-old man presents with insidious iron-deficiency anaemia. Colonoscopy reveals a polypoid, fungating mass in the caecum. This presentation is typical of carcinoma arising in the:

- A. Left colon
- B. Rectum
- C. Duodenum
- D. Right colon
- E. Ileum

3. A patient with metastatic midgut neuroendocrine tumor develops flushing, diarrhea, and echocardiographic evidence of tricuspid stenosis. This carcinoid syndrome occurs because:

- A. The primary tumor secretes gastrin directly into the systemic circulation
- B. The tumor has transformed into an adenocarcinoma
- C. Liver metastases release serotonin directly into the systemic circulation, bypassing hepatic inactivation
- D. GIST has co-developed alongside the NET
- E. The tumor has invaded the pancreas

4. A 55-year-old presents with a gastric mass. Biopsy shows spindle cells, and immunohistochemistry suggests origin from the interstitial cells of Cajal. This tumor is:

- A. Neuroendocrine tumor
- B. Adenocarcinoma
- C. Lymphoma
- D. Leiomyoma of the esophagus
- E. Gastrointestinal stromal tumor (GIST)

5. Which dietary factor is proposed to increase colorectal carcinoma risk via decreased stool bulk and altered gut microbiota?

- A. High intake of vitamin C
- B. Low intake of unabsorbable vegetable fiber
- C. High intake of vitamin E
- D. Regular aspirin use
- E. High fiber intake

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — FAP is caused by mutation of the APC tumor suppressor gene on chromosome 5, which regulates proliferation via WNT signaling.

2 • D — Right colon carcinomas tend to be polypoid, fungating masses that present with insidious anaemia, unlike left colon (napkin-ring obstruction) or rectal (bleeding/tenesmus) tumors.

3 • C — Carcinoid syndrome typically requires liver metastases, since normally the liver inactivates portal-vein serotonin to 5-HIAA before it reaches the systemic circulation.

4 • E — GIST arises from the interstitial cells of Cajal (the GIT pacemaker cells) and is most common in the stomach; most are spindle cell lesions.

5 • B — Low fiber/high refined carbohydrate intake decreases stool bulk and alters gut microbiota composition, increasing toxic bacterial oxidative by-products.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL3, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.